Parallel lines and transversals

Core Concepts: Parallel lines and transversals

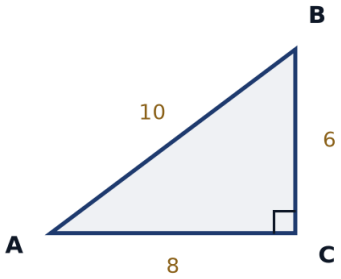
1 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93°

b. 103°

c. 85°

d. 138°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 12

c. 10

d. 12

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 86°

b. 96°

c. 78°

d. 138°

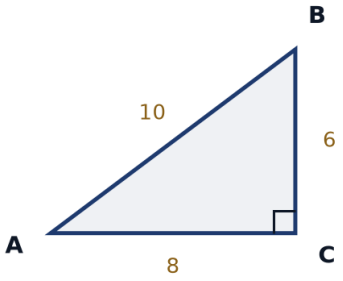
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 10

b. 14

c. 12

d. 14

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 79°

b. 89°

c. 71°

d. 138°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

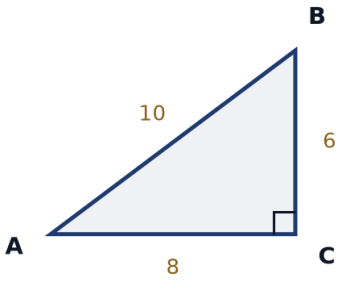
9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 12

b. 16

c. 14

d. 16



Exam-Style Problem Solving & Modeling

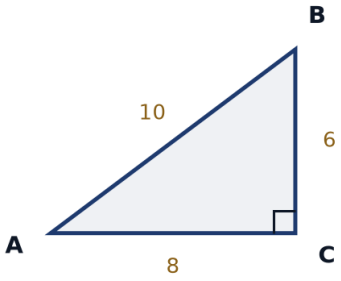
10 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 72°

b. 82°

c. 64°

d. 138°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 14

b. 18

c. 16

d. 18

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